

JEE MAINS GRAND TEST SERIES



NARAYANA
IIT ACADEMY
INDIA

40+
YEARS
OF EXCELLENCE

Sec: SR.IIT_*CO-SC(MODEL-A,B&C)

Time: 3HRS

Date:24-12-23

Max. Marks: 300

Name of the Student: _____

H.T. NO:

24-12-23_SR.STAR CO-SUPER CHAINA(MODEL-A,B&C)_JEE-MAIN_GTM-2_SYLLABUS

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

MATHEMATICS: **TOTAL SYLLABUS**

SUBJECT	MISTAKES		
	JEE SYLLABUS Q'S	JEE EXTRA SYLLABUS Q'S	TOTAL Q'S
MATHS			
PHYISCS			
CHEMISTRY			

For More Material Join: @JEEAdvanced_2025

PHYSICS

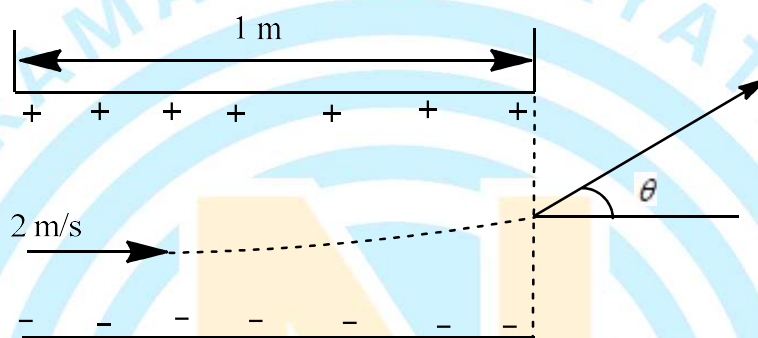
MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A uniform electric field $E = (8 \text{ m/e}) \text{ V/m}$ is created between two parallel plates of length 1m as shown in the figure (where $m = \text{mass of electron}$ and $e = \text{charge of electron}$). An electron enters the field symmetrically between the plates with a speed of 2 m/s. The angle of deviation (θ) of the path of the electron as it comes out of the field will be _____



- A) $\tan^{-1}(4)$ B) $\tan^{-1}(2)$ C) $\tan^{-1}(1/3)$ D) $\tan^{-1}(3)$
2. Four point charges equal to $-Q$ are placed at four corners of a square and a point charge q is at its centre. If the system is in equilibrium the value of q is
- A) $-\frac{Q}{4}(1+2\sqrt{2})$ B) $\frac{Q}{4}(1+2\sqrt{2})$ C) $-\frac{Q}{2}(1+2\sqrt{2})$ D) $\frac{Q}{2}(1+2\sqrt{2})$
3. A horizontal overhead power line is at a height of 4 m from the ground and carries a current of 100 A from east to west. The magnetic field directly below it on the ground is ($\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$)
- A) $2.5 \times 10^{-7} \text{ T}$ northwards B) $2.5 \times 10^{-7} \text{ T}$ southwards
C) $5 \times 10^{-6} \text{ T}$ northwards D) $5 \times 10^{-6} \text{ T}$ southwards
4. A magnetic needle lying parallel to a magnetic field requires W units of work to turn through 60° . The torque needed to maintain the needle in this position will be
- A) $\sqrt{3}W$ B) W C) $\frac{\sqrt{3}}{2}W$ D) $2W$

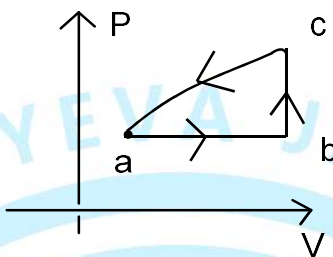
5. A metallic rod of length l is tied to a string of length $2l$ and made to rotate with angular speed ω on a horizontal table with one end of the string fixed. If there is a vertical magnetic field 'B' in the region, the emf induced across the ends of the rod is
- A) $\frac{5}{2}.B\omega l^2$ B) $\frac{2B\omega l^2}{2}$ C) $\frac{3}{2}.B\omega l^2$ D) $\frac{4}{2}.B\omega l^2$
6. A sinusoidal voltage of peak value 283V and angular frequency of 320rad/s is applied to a series LCR circuit. Given that resistance $R = 5\Omega$, inductance $L = 25 \text{ mH}$ and capacitance $C = 1000\mu F$. The total impedance and phase difference between the voltage across the source and the current will nearly be (respectively)
- A) 10Ω and $\tan^{-1}\left(\frac{5}{3}\right)$ B) 10Ω and $\tan^{-1}\left(\frac{8}{3}\right)$
C) 7Ω and $\tan^{-1}\left(\frac{5}{3}\right)$ D) 7Ω and 45°
7. The magnetic field of a plane electromagnetic wave is given by $\vec{B} = 3 \times 10^{-8} [\sin(200\pi(y+ct))] \hat{i} \text{ T}$. Where $C = 3 \times 10^8 \text{ m/s}$ is the speed of light. The corresponding electric field is
- A) $\vec{E} = 9 \sin[200\pi(y+ct)] \hat{k} \text{ v/m}$
B) $\vec{E} = -10^{-6} \sin[\sin(200\pi)(y+ct)] \hat{k} \text{ v/m}$
C) $\vec{E} = 3 \times 10^{-8} [\sin(200\pi(y+ct))] \hat{k} \text{ v/m}$
D) $\vec{E} = -9 \sin[200\pi(y+ct)] \hat{k} \text{ v/m}$
8. Two coherent sources of light interfere. The intensity ratio of two sources is 1:4. For this interference pattern if the value of $\frac{I_{\max} + I_{\min}}{I_{\max} - I_{\min}} =$
- $\left(\begin{array}{l} I_{\max} = \text{Maximum intensity value} \\ I_{\min} = \text{Minimum intensity value} \end{array} \right)$
- A) 1.5 B) 1.25 C) 0.5 D) 1

9. The energy required for the electron excitation in Li^{++} from the first to the third Bohr orbit is
- A) 12.1 eV B) 36.3 eV C) 108.8 eV D) 122.4 eV
10. **STATEMENT – 1:** Energy is released in nuclear fission.
- STATEMENT – 2:** Total binding energy of the fission fragments is larger than the total binding energy of parent nucleus.
- A) Statement – 1 is True, Statement – 2 is True; Statement – 2 is a correct explanation for Statement – 1.
- B) Statement – 1 is True, Statement – 2 is True; Statement – 2 is **NOT** a correct explanation for Statement – 1.
- C) Statement – 1 is True, Statement – 2 is False.
- D) Statement – 1 is False, Statement – 2 is True.
11. In a Vernier callipers, each cm on the main scale is divided into 20 equal parts. 10 Vernier scale divisions are equivalent to 9 main scale divisions. The value of Vernier constant will be $\text{_____} \times 10^{-2} \text{ mm}$
- A) 5 B) 50 C) 500 D) 200
12. A boy can throw a stone up to a maximum height of 10m. The maximum horizontal distance that the boy can throw the same stone up to will be _____ (in m)
- A) 10 B) $10\sqrt{2}$ C) 20 D) $20\sqrt{2}$
13. Moment of inertia of a square plate of side l about the axis passing through one of the corners and perpendicular to the plane of square plate is given by
- A) $\frac{Ml^2}{6}$ B) $\frac{2}{3}Ml^2$ C) Ml^2 D) $\frac{Ml^2}{12}$
14. Minimum energy required to move a satellite of mass m from an orbit of radius $2R$ to $3R$ is, M = Mass of the planet
- A) $\frac{GMm}{12R}$ B) $\frac{GMm}{3R}$ C) $\frac{GMm}{8R}$ D) $\frac{GMm}{6R}$

15. The bulk modulus of a liquid is $3 \times 10^{10} \text{ N/m}^2$. The pressure required to reduce the volume of liquid by 2% is

- A) $3 \times 10^8 \text{ N/m}^2$ B) $9 \times 10^8 \text{ N/m}^2$ C) $6 \times 10^8 \text{ N/m}^2$ D) $12 \times 10^8 \text{ N/m}^2$

16. A sample of an ideal gas is taken through cyclic process abca as shown in the figure. The change in internal energy of the gas along the path ca is -180J. The gas absorbs 250J of heat along the path ab and 60J along the path bc. The work done by the gas along the path abc is



- A) 120J B) 130J C) 140J D) 100J

17. The ratio of specific heats $\left(\frac{C_p}{C_v} = \gamma\right)$ in terms of degree of freedom (f) is

(C_p = Specific heat capacity at constant Pressure
 C_v = Specific heat capacity at constant Volume)

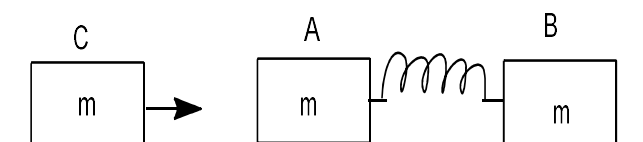
- A) $1 + \frac{f}{3}$ B) $1 + \frac{2}{f}$ C) $1 + \frac{f}{2}$ D) $1 + \frac{1}{f}$

18. A simple harmonic motion is represented by $y = 5[\sin(3\pi t) + \sqrt{3} \cos(3\pi t)] \text{ cm}$

The amplitude and time period of the motion are

- A) $5\text{cm}, \frac{3}{2}\text{S}$ B) $10\text{cm}, \frac{2}{3}\text{S}$ C) $5\text{cm}, \frac{2}{3}\text{S}$ D) $10\text{cm}, \frac{3}{2}\text{S}$

19. Two identical blocks A and B each of mass m resting on the smooth horizontal floor are connected by a light spring of natural length L and spring constant k . A third block c of mass m moving with a speed v along the line joining A and B collides elastically with A. The maximum compression in the spring is



- A) $v\sqrt{\frac{m}{2k}}$ B) $\sqrt{\frac{mv}{2k}}$ C) $\sqrt{\frac{m}{2k}}$ D) $\sqrt{\frac{mv}{k}}$

20. A large number of water drops each of radius r combine to form a drop of radius R . If the surface tension is T and mechanical equivalent of heat is J , the rise in heat energy per unit volume will be

A) $\frac{3T}{J}$ B) $\frac{2T}{rJ}$ C) $\frac{3T}{J}\left(\frac{1}{r} - \frac{1}{R}\right)$ D) $\frac{2T}{J}\left(\frac{1}{r} - \frac{1}{R}\right)$

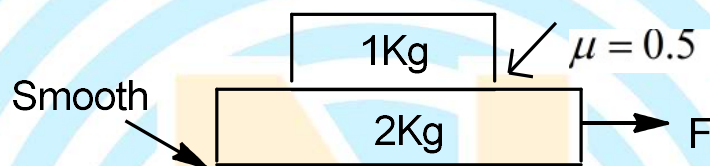
SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

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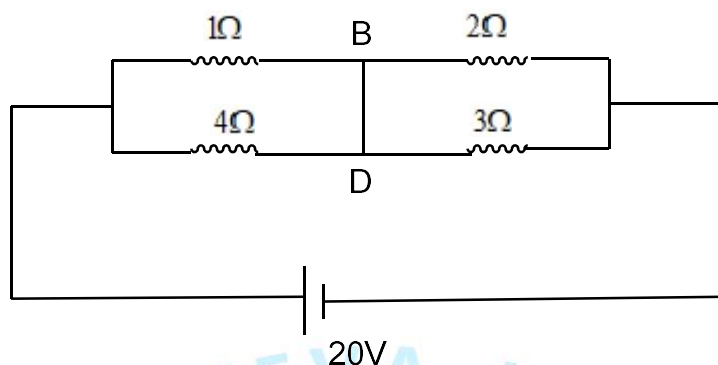
Marking scheme: +4 for correct answer, -1 in all other cases.

21. The coefficient of static friction between two blocks is 0.5 and table is smooth. The maximum horizontal force that can be applied to move the block together is _____ N.

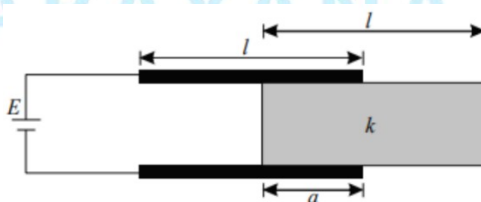


22. The centre of mass of a solid hemisphere of radius 8cm is x cm from the centre of the flat surface. Then value of x is _____.
23. The centre of a wheel pure rolling on a plane surface moves with a speed of v_0 . A particle on the rim of the wheel at the same level as the centre will be moving with $\sqrt{x}v_0$. The value of x is _____.
24. An ideal gas is expanding such that $PT^2 = C$. If gas expands from 0K, the co-efficient of volume expansion of the gas is $\frac{x}{T}$. The value of x is _____.
25. A transverse wave is described by the equation $y = y_0 \sin 2\pi \left[ft - \frac{x}{\lambda} \right]$. The maximum particle velocity is equal to four time the wave velocity if $\lambda = \frac{\alpha y_0}{2}$. The integer part of α is _____.(to the nearest integer)

26. In the given circuit diagram, a wire is joining point B and D. The current in this wire is _____ A.



27. A prism of refractive index μ and angle of prism A is placed in position of minimum angle of deviation. If minimum angle of deviation is also A, then in terms of refractive index the value of A is $\alpha \cos^{-1}\left(\frac{\mu}{\beta}\right)$. The value of $\alpha + \beta$ is _____.
28. An α - particle and a proton are accelerated from rest by a potential difference of 100 v. After this their de Broglie's wavelength are λ_α and λ_p . The ratio $\frac{\lambda_p}{\lambda_\alpha}$ to nearest integer is _____.
29. Consider the situation shown in figure. The plates of the capacitor have plate area A and are clamped in the laboratory. A dielectric slab is released from rest with length a inside the capacitor. Neglecting any effect of friction or gravity the time period of periodic motion of slab will be $T = x \left(\sqrt{\frac{(l-a)lmd}{\epsilon_0 AE^2 (K-1)}} \right)$, Then $x =$ _____. (K is dielectric constant of the slab.)



30. A particle is moving in a circular path of radius a under the action of an attractive potential $U = -\frac{K}{2r^2}$. Its total energy is _____.

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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31. Which of the following statement is incorrect ?

A) The order of the ionic character of the following chlorides is



B) The reducing nature of alkaline earth metals in aqueous medium follows the increasing order as $Be < Mg < Ba < Ca < Sr$

C) Beryllium is not readily attacked by acid because of the presence of an oxide film on the surface of Be.

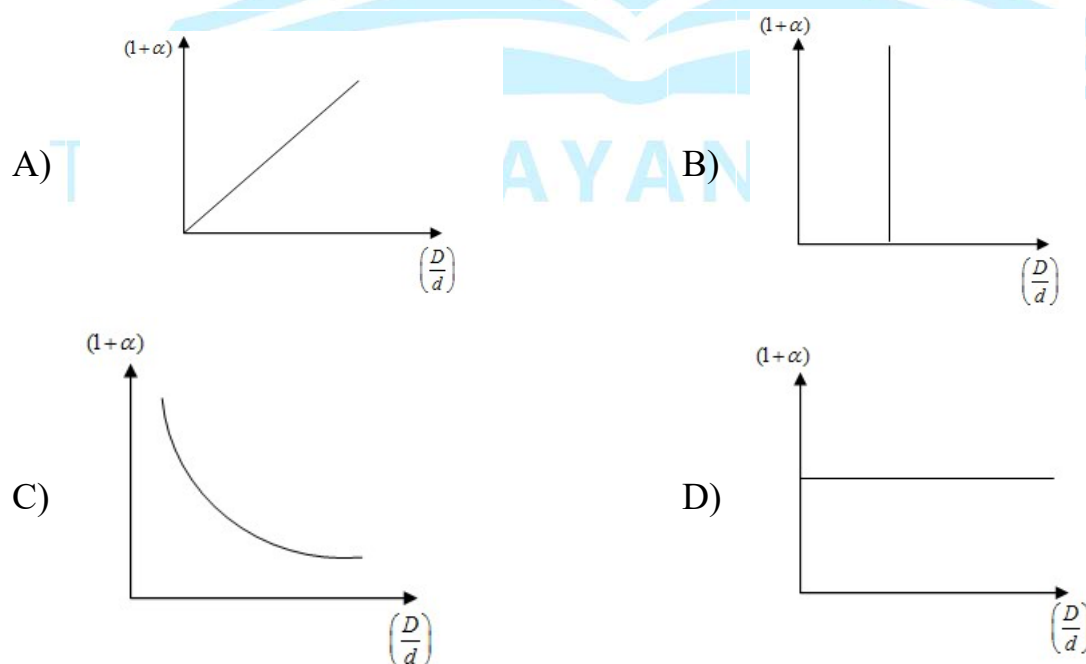
D) Sodium hydroxide is produced on a large scale by the electrolysis of aqueous solution of $NaCl$ using Hg as cathode

32. Which of the following product in a hydrogen atom, dependent of the principal quantum number 'n' ? [Given : E_n = Total energy of electron; r_n = Radius of Bohr orbit ; v_n = velocity of electron]

- A) $v_n^2 \cdot r_n$ B) $E_n \cdot r_n$ C) $\left(\frac{1}{E_n}\right) \cdot v_n^2$ D) $v_n \cdot \left[\frac{1}{r_n}\right]$

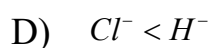
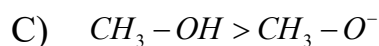
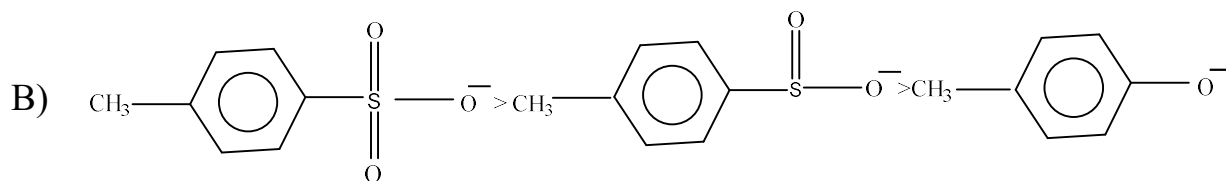
33. At a given temperature the dissociation equilibrium of $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$, the variation of $(1+\alpha)$ against (D/d) is represented on :

(Given : α = degree of dissociation, D = initial vapour density, d = vapour density at equilibrium)



34. Find the ΔpH (initial pH – final pH) when 100 mL of 0.01M HCl is added in a solution containing 0.1 m moles of $NaHCO_3$ solution of negligible volume
($ka_1 = 10^{-7}$, $ka_2 = 10^{-11}$ for H_2CO_3)
- A) $6 + 2 \log 3$ B) $6 - \log 3$ C) $6 + 2 \log 2$ D) $6 - 2 \log 3$
35. For an ideal gas having molar mass M, specific heat at constant pressure can be given as:
- A) $\frac{\gamma RM}{(\gamma - 1)}$ B) $\frac{R}{M(\gamma - 1)}$ C) $\frac{RM}{(\gamma - 1)}$ D) $\frac{\gamma R}{M(\gamma - 1)}$
36. For the reaction $2H_2 + 2NO \rightleftharpoons N_2 + 2H_2O$ the following mechanism has been suggested :
- $2NO \rightleftharpoons N_2O_2$ [equilibrium constant = K_1 (fast)]
- $N_2O_2 + H_2 \xrightarrow{K_2} N_2O + H_2O$ (slow)
- $N_2O + H_2 \xrightarrow{K_3} N_2 + H_2O$ (fast)
- According to following mechanism is correctly matched.
- A) $-\frac{1}{2} \frac{d[H_2]}{dt} = K[NO][H_2]$ where $K = K_2 \times K_3$
- B) $-\frac{1}{2} \frac{d[NO]}{dt} = K[NO][H_2]$ where $K = K_2 \times K_1$
- C) $\frac{d[N_2]}{dt} = K[NO][H_2]$ where $K = K_2 \times K_1$
- D) $-\frac{1}{2} \frac{d[NO]}{dt} = K[NO]^2 [H_2]$ where $K = K_2 \times K_1$
37. Resistance of 0.2 M solution of an electrolyte is 50Ω the specific conductance of the solution is $1.4 Sm^{-1}$ the resistance of 0.5 M solution of the same electrolyte is 280Ω the molar conductivity of 0.5 M solution of the electrolyte in $Sm^2 mol^{-1}$ is
- A) 5×10^{-3} B) 5×10^3 C) 5×10^2 D) 5×10^{-4}

38. Which of the following order of leaving group is incorrect ?



39. Statement-1: The 5th period of periodic table contains 18 elements not 32.

Statement-2: The order in which the energy of available orbitals 4d, 5s and 5p increases is $5s < 4d < 5p$ and the total number of orbitals available are 9 and thus 18 electrons can be accommodated.

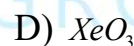
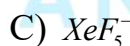
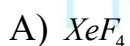
A) Statement-1 is true, statement-2 is true, statement-2 is a correct explanation for statement-1.

B) Statement-1 is true, statement-2 is true, statement-2 is not a correct explanation for statement-1.

C) Statement-1 is true, statement-2 is false

D) Statement-1 is false, statement-2 is true

40. Which of the following compound used maximum number of non-axial d-orbitals in hybridization of central atom ?



41. Which of the following option is CORRECT ?

A) $3p-3p > 3p-3d > 3d-3d$ (π bond strength)

B) Ethanol < Glycerol < ethylene glycol (Viscosity)

C) $\text{XeO}_3\text{F}_2 > \text{XeF}_4 = \text{XeF}_2$ (Dipole moment)

D) In 13th group Al has least S.R.P

42. Assertion (A) : 1 mole of FeC_2O_4 is oxidised by 0.6 mole of MnO_4^- in acidic medium

Reason (R) : MnO_4^- oxidises both Fe^{2+} as well as $C_2O_4^{2-}$

A) Both A and R are true and R is the correct explanation of A

B) Both A and R are true but R is not the correct explanation of A

C) A is true but R is false

D) A is false but R is true

43. Which one amongst the following are good oxidizing agents ?

a) Sm^{2+}

b) Ce^{2+}

c) Ce^{4+}

d) Tb^{4+}

Choose the most appropriate answer from the options given below:

A) c only

B) d only

C) a and b only

D) c and d only

44. The IUPAC name dichloride bis (ethane 1,2-di amine) cobalt (III) chloride referred for

A) $[Co(dien)_2Cl_2]Cl$

B) $[Co(dien)Cl_2]Cl$

C) $[Co(en)_2Cl_2]Cl$

D) $[Co(en)_2Cl]Cl_2$

45. In Dumas method for estimation of nitrogen, 0.3 g of an organic compound gave 50 mL of nitrogen collected at 300K temperature and 715 mm pressure. Calculate the percentage composition of nitrogen in the compound. (Aqueous tension at 300K = 15mm)

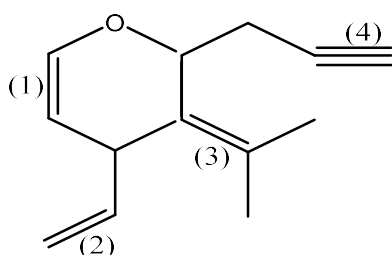
A) 40%

B) 70%

C) 17.5%

D) 6%

46. The correct reactivity order of following $C=C/C\equiv C$ bonds towards H^+ is



A) $4 > 3 > 2 > 1$

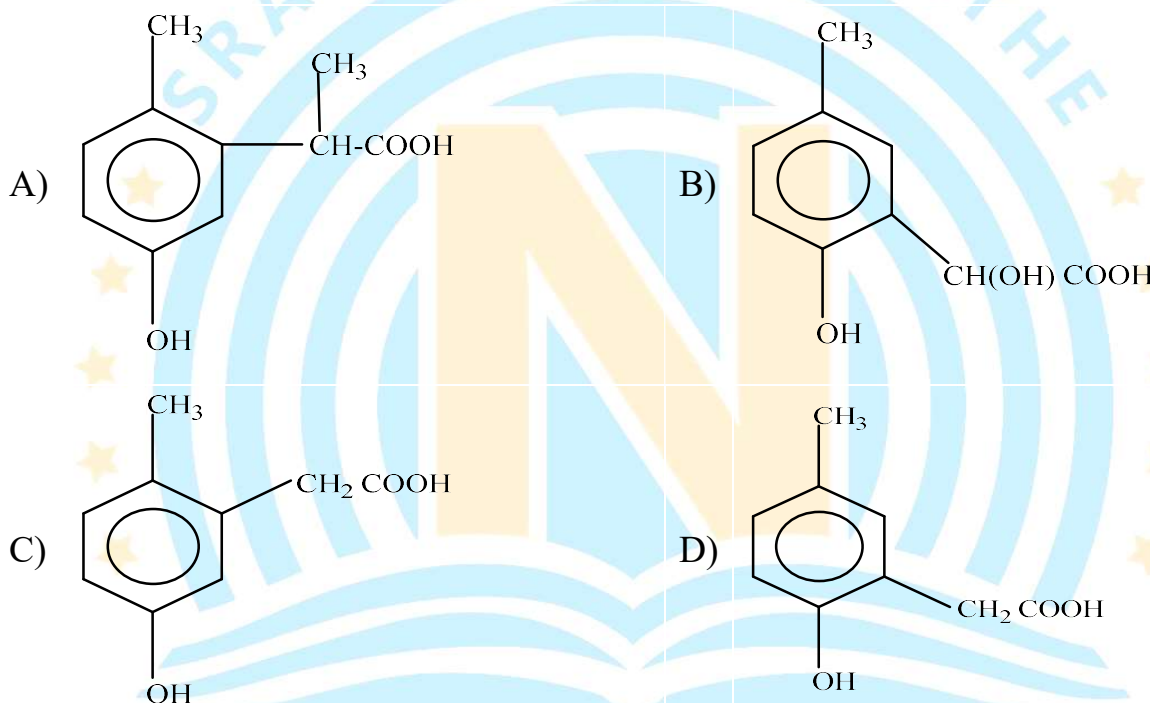
B) $3 > 2 > 1 > 4$

C) $1 > 3 > 4 > 2$

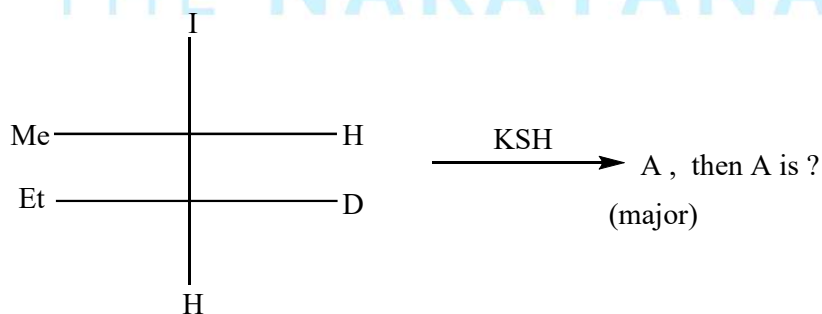
D) $1 > 3 > 2 > 4$

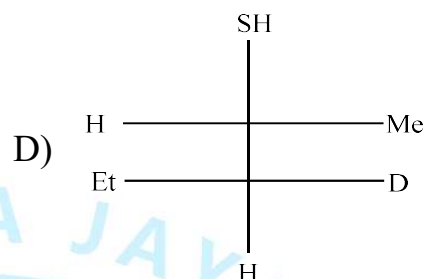
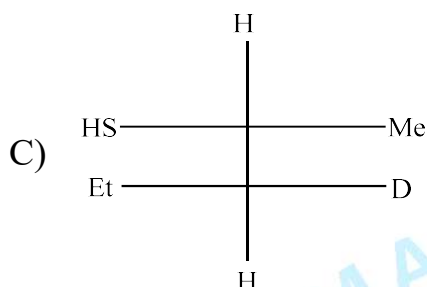
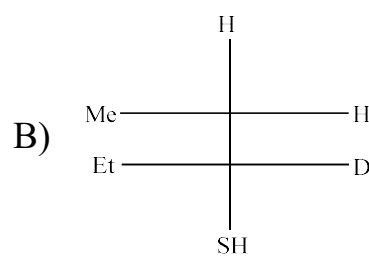
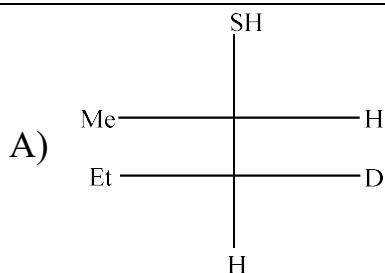
47. Test	Reagent
1) Lucas test	a) $C_6H_5SO_2Cl$
2) Hinsberg test	b) Alc $KOH + CHCl_3$
3) Iso cyanide test	c) An hyd $ZnCl_2 + Conc.HCl$
4) Molisch test	d) Alc α -naphthol+Conc. H_2SO_4
A. 1-c, 2-a, 3-b, 4-d	B) 1-a, 2-c, 3-b, 4-d
C. 1-c, 2-a, 3-d, 4-b	D) 1-c, 2-b, 3-a, 4-d

48. P-Cresol reacts with chloroform in alkaline medium to give the compound A which adds HCN to form, the compound B. The latter on acidic hydrolysis gives chiral carboxylic acid. The structure of carboxylic acid is

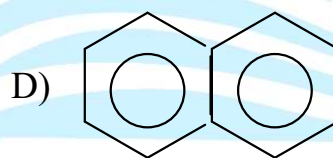
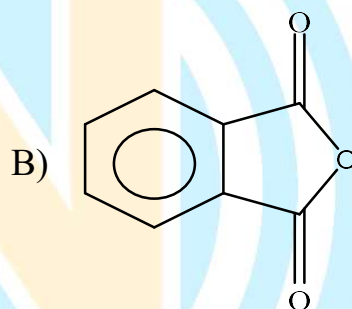
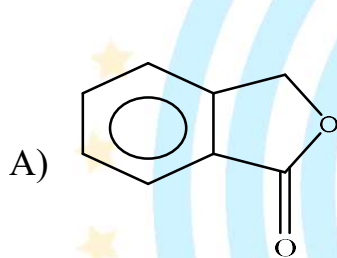


49.





50.



SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

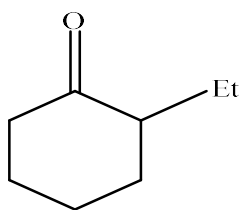
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51. 500 mL of 2M impure H_2SO_4 sample can completely react with 1.5 L of 1 M NaOH solution. Then find ' α ' ?

Where $\alpha = \frac{\text{Weight of } \text{H}_2\text{SO}_4}{25}$.

52. 1 gram of arsenic dissolved in 86 gm of benzene brings down the freezing point to 5.31°C from 5.50°C . If k_f of benzene is $4.9 \text{ k. molality}^{-1}$, then atomicity of molecule is ?
[Given atomic weight of As = 75 amu]
53. How many of the compound will evolve methane gas on treatment with CH_3MgBr
- (A) H_2O (B) CH_3-OH (C) CH_3COCH_3 (D) CH_3COOH
(E) CH_3CONH_2 (F) CH_3COCl (G) $\text{CH}_3\text{COOCH}_3$ (H) CH_3Cl
(I) $\text{Ph}-\text{OH}$
54. $\text{C}_6\text{H}_{12} \xrightarrow[\text{CCl}_4]{\text{Br}_2} \text{C}_6\text{H}_{12}\text{Br}_2$
X (Optically inactive single product with 2 chiral center)
How many structures of "X" can satisfy above reaction condition ?
55. How many of the following ions gives precipitate with $\text{H}_2\text{S}+\text{NH}_4\text{OH}$
 $\text{Pb}^{+2}, \text{Cu}^{+2}, \text{Al}^{+3}, \text{Ca}^{+2}, \text{Mn}^{+2}, \text{Zn}^{+2}, \text{Cd}^{+2}, \text{Cr}^{+3}, \text{Mg}^{+2}$
56. How many isomers obtained by the halogenation of ethane?
57. The number of monobromo products (including stereoisomers) obtained during heating of methylcyclobutane with Br_2 .
58. The condensation of two amino acids, glycine and ($\pm$) alanine yields total products x.
Find value of x. Assume that only dipeptides are formed.
59. How many of the following compounds contains carboxylic acid group ? Ascorbic acid, Phthalic acid, Cinnamic acid, acrylic acid, aspirin, formic acid, picric acid, adipic acid.
60. Total number of β -keto monocarboxylic acids (including stereoisomers) which on



heating give .

MATHEMATICS

MAX.MARKS: 100

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This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

61. The set of values of a for which all extremum of function $f(x) = x^3 - 3ax^2 + 3(a^2 - 1)x + 1$ lie in the interval $(-2, 4)$ is
- A) $(3, 4)$ B) $(-1, 3)$ C) $(-3, -1)$ D) $(-2, -1)$
62. $\int [\sin(101x) \cdot \sin^{99} x] dx$ equals
- A) $\frac{\sin(100x)(\sin x)^{100}}{100} + C$ B) $\frac{\cos(100x)(\sin x)^{100}}{100} + C$
- C) $\frac{\cos(100x)(\cos x)^{100}}{100} + C$ D) $\frac{\sin(100x)(\sin x)^{101}}{101} + C$
63. The function $f(x) = \begin{cases} \left(\frac{4}{3}\right)^{\frac{\tan 4x}{\tan 3x}} & ; 0 < x < \frac{\pi}{2} \\ b + 2 & ; x = \frac{\pi}{2} \\ (1 + |\cos x|)^{\frac{a}{b}|\tan x|} & ; \frac{\pi}{2} < x < \pi \end{cases}$ then the sum $(a + b)$ if f is continuous at $x = \frac{\pi}{2}$ is
- A) -1 B) 1 C) 2 D) -2
64. Let $f(x) = ([a]^2 - 5[a] + 4)x^3 - (6\{a\}^2 - 5\{a\} + 1)x - (\tan x)\operatorname{sgn} x$, be an even function for all $x \in \mathbb{R} - (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$ then sum of all possible values of a is (where $[.]$ and $\{.\}$ denote the greatest integer function and fractional part functions respectively)
- A) $\frac{70}{3}$ B) $\frac{35}{6}$ C) $\frac{35}{3}$ D) $\frac{70}{6}$

65. Let $y = y(x)$ be the solution of the differential equation

$$\frac{dy}{dx} = 1 + xe^{y-x}, -\sqrt{2} < x < \sqrt{2}, y(0) = 0 \text{ then the minimum value of } y(x), x \in (-\sqrt{2}, \sqrt{2}) \text{ is}$$

equal to

A) $(2 - \sqrt{3}) - \log_e 2$

B) $(1 - \sqrt{3}) - \log_e (\sqrt{3} - 1)$

C) $(1 + \sqrt{3}) - \log_e (\sqrt{3} - 1)$

D) $(2 + \sqrt{3}) + \log_e 2$

66. If $P = \begin{pmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$, $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $Q = PAP^T$ then the inverse of $P^T Q^{2023} P$ is

A) $\begin{pmatrix} 1 & 2023 \\ 0 & 1 \end{pmatrix}$

B) $\begin{pmatrix} 2023 & 0 \\ 1 & 1 \end{pmatrix}$

C) $\begin{pmatrix} 1 & 0 \\ 1 & -2023 \end{pmatrix}$

D) $\begin{pmatrix} 1 & -2023 \\ 0 & 1 \end{pmatrix}$

67. The number of numbers between 2000 and 5000 that can be formed with the digits 0,1,2,3,4 (repetition of digits is not allowed) and are multiple of 3 is :

A) 24

B) 30

C) 36

D) 48

68. The range of the function $f(x) = \sqrt{\cos^{-1}(\sqrt{\log_4 x}) - \frac{\pi}{2}} + \sin^{-1}\left(\frac{1+x^2}{4x}\right)$ is equal to

A) $\left(0, \frac{\pi}{2} + \sqrt{\frac{\pi}{2}}\right)$

B) $\left[\frac{\pi}{2}, \frac{\pi}{2} + \sqrt{\frac{\pi}{2}}\right]$

C) $\left[\frac{\pi}{6}, \frac{\pi}{2}\right]$

D) $\left\{\frac{\pi}{6}\right\}$

69. If $f(x) = \begin{cases} -e^{-x} + k, & x \leq 0 \\ e^x + 1, & 0 < x < 1 \\ ex^2 + \lambda, & x \geq 1 \end{cases}$ is one-one and monotonically increasing for all

$x \in \mathbb{R}$, then difference of maximum value of k and minimum value of λ is

A) 0

B) 1

C) 2

D) 3

70. $f(x)$ is differentiable function which satisfies the equation

$$f(x) = -\int_0^x f(t) \tan t dt + \int_0^x \tan(t-x) dt \text{ where } x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \text{ passes through } (0,0), \text{ then}$$

the maximum value of $f(x)$ is

- A) 1 B) -1 C) 0 D) 2

71. Four digit numbers are formed using the digits from the set $\{0,1,2,3,4,5\}$ repetition of digits is allowed then

Statement- S_1 The number of such numbers formed that are odd is 480

Statement- S_2 The number of such numbers formed such that it contains exactly three different digits is 360

- A) S_1, S_2 are true B) S_1, S_2 are false
C) S_1 is true and S_2 is false D) S_1 is false and S_2 is true

72. The shortest distance between the lines $\frac{x-3}{2} = \frac{y-2}{3} = \frac{z-1}{-1}$ and $\frac{x+3}{2} = \frac{y-6}{1} = \frac{z-5}{3}$ is

- A) $\frac{18}{\sqrt{5}}$ B) $\frac{22}{3\sqrt{5}}$ C) $\frac{46}{3\sqrt{5}}$ D) $6\sqrt{3}$

73. Two loaded dice each have the property that 2 or 4 is three times as likely to appear as 1 or 3 or 5 or 6 on each roll. When two such dice are rolled, the probability of obtaining a total of 7 is

- A) $\frac{1}{8}$ B) $\frac{1}{7}$ C) $\frac{7}{50}$ D) none of these

74. Circles of radii 36 and 9 touch externally. The radius of the circle which touches the two circles externally and also their common tangent can be

- A) 4 B) 5 C) $\sqrt{17}$ D) $\sqrt{18}$

75. Consider a branch of the hyperbola $x^2 - 2y^2 - 2\sqrt{2}x - 4\sqrt{2}y - 6 = 0$ with vertex at the point A. Let B be one of the end points of its latus rectum and C be the focus of the hyperbola nearest to the point A, the area of the triangle ABC is :
- A) $\sqrt{\frac{3}{2}}$ B) $\sqrt{\frac{3}{2}} - 1$ C) $\sqrt{\frac{3}{2}} + 1$ D) $\sqrt{\frac{2}{3}}$
76. The vector $\vec{r}$, which is normal to both $\vec{a} = 4\hat{i} + 5\hat{j} - \hat{k}$ and $\vec{b} = \hat{i} - 4\hat{j} + 5\hat{k}$ and $\vec{r} \cdot \vec{c} = 21$ Where $\vec{c} = 3\hat{i} + \hat{j} - \hat{k}$ is _____
- A) $7(\hat{i} + \hat{j} - \hat{k})$ B) $7(-\hat{i} + \hat{j} - \hat{k})$ C) $7(\hat{i} - \hat{j} - \hat{k})$ D) $7(2\hat{i} + \hat{j} + 2\hat{k})$
77. Let A and B be two independent events such that $P(A) = \frac{1}{3}$ and $P(B) = \frac{1}{6}$. Then, which of the following is TRUE?
- A) $P\left(\frac{A}{B}\right) = \frac{2}{3}$ B) $P\left(\frac{A'}{B'}\right) = \frac{1}{3}$ C) $P\left(\frac{A}{B'}\right) = \frac{1}{3}$ D) $P\left(\frac{A}{A \cup B}\right) = \frac{1}{4}$
78. Let α, β be two roots of the equation $x^2 + (20)^{1/4}x + (5)^{1/2} = 0$. Then $\alpha^8 + \beta^8$ is equal to
- A) 10 B) 50 C) 5 D) 30
79. Let $A = \{1, a_1, a_2, \dots, a_{18}, 77\}$ be a set of integers with $1 < a_1 < a_2 < \dots < a_{18} < 77$. Let the set $A + A = \{x + y : y \in A\}$ contain exactly 39 elements. Then, the value of $a_1 + a_2 + \dots + a_{18}$ is equal to _____.
- A) 600 B) 702 C) 800 D) 200
80. Normals of parabola $y^2 = 4x$ at P and Q meets at $R(x_2, 0)$ and tangents at P and Q meets at $T(x_1, 0)$. If $x_2 = 3$, then the area of quadrilateral PTQR is _____
- A) 4 B) 8 C) 12 D) 6

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

81. Let $p, q, r \in \mathbb{R}^+$ and $27pqr \geq (p+q+r)^3$ and $3p+4q+5r=12$ then the value of $8p+4q-7r$ is equal to _____.
82. If $\lim_{x \rightarrow 1} \frac{\sin(3x^2 - 4x + 1) - x^2 + 1}{2x^3 - 7x^2 + ax + b} = -2$, then the value of $(a-b)$ is equal to _____
83. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$, $x \in \mathbb{R}$. Then $|f(2)|$ is equal to _____
84. Suppose a class has 7 students. The average marks of these students in the mathematics examination is 62, and their variance is 20. A student fails in the examination if he/she gets less than 50 marks, then in worst case, the number of students that can fail is _____
85. Consider the set of all triangles OPQ where O is the origin and P and Q are distinct points in the plane with non-negative integral coordinates (x, y) such that $5x + y = 99$. Number of such distinct triangles whose area is a positive integer is _____
86. If the area of the region bounded by the curves $y = ex \log x$ and $y = \frac{\log x}{ex}$ is $\frac{K}{4e}$, find the value of $[K]$ (Where $[\cdot]$ represents greatest integer function) _____
87. Using all of 0,0,0,1,1,1,-1,-1,-1, a set 'S' of all 3×3 matrices formed. If number of symmetric matrices in S is K and number of matrices having trace zero in S is P, then the value of $\frac{P}{10} + K$ is _____
88. The foot of perpendicular of the point $(0, 2, 7)$ on the line $\frac{x+2}{-1} = \frac{y-1}{3} = \frac{z-3}{-2}$ is (α, β, γ) then the value of $\frac{4\alpha\beta}{\gamma}$ is _____

89. A spherical iron ball of radius 10cm is coated with a layer of ice of uniform thickness that melts at a rate of $50 \text{ cm}^3/\text{min}$. When the thickness of the ice is 5cm, the rate at which the thickness (in cm/min) of the ice decreases is $\frac{1}{k\pi}$, find the value of k.
90. If the equation $z^4 + a_1z^3 + a_2z^2 + a_3z + a_4 = 0$ where a_1, a_2, a_3, a_4 are real coefficient different from zero has purely imaginary roots then find the value of the expression $\frac{a_3}{a_1a_2} + \frac{a_1a_4}{a_2a_3}$.

